

Reproductive aging and menopause-like transition in the menstruating spiny mouse (*Acomys cahirinus*)

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STUDY QUESTION: Does the naturally menstruating spiny mouse go through menopause?

SUMMARY ANSWER: Our study is the first to show a natural and gradual menopausal transition in a rodent.

WHAT IS KNOWN ALREADY: Age-related depletion of the human ovarian reserve (OvR) leads to menopause, the permanent cessation of menstruation and reproduction. Current rodent models of menopause are inappropriate for inferences of the human condition, as reproductive senescence is abrupt or induced through ovariectomy. The spiny mouse is the only confirmed rodent with a naturally occurring menstrual cycle.

STUDY DESIGN, SIZE, DURATION: Histological assessment of virgin spiny mice occurred in females aged 6 months ($n = 14$), 1 year ($n = 7$), 2 years ($n = 13$), 3 years ($n = 9$) and 4 years ($n = 9$). Endocrinology was assessed in a further 9 females per age group. Five animals per group were used for ovarian stereology with additional ovaries collected at prenatal Day 35 ($n = 3$), day of birth ($n = 5$), postnatal Days 35 ($n = 5$) and 100 ($n = 5$) and 15 months ($n = 5$).

PARTICIPANTS/MATERIALS, SETTING, METHODS: Morphological changes in the reproductive system were examined using hematoxylin and eosin stains. Proliferating cell nuclear antigen immunohistochemistry assessed endometrial proliferation and sex steroids estradiol and testosterone were assayed using commercial ELISA kits.

MAIN RESULTS AND THE ROLE OF CHANCE: The proportion of females actively cycling was 86% at 6 months, 71% at 1 year, 69% at 2 years, 56% at 3 years and 44% at 4 years. Uterine and ovarian weights declined steadily from 1 year in all groups and corresponded with loss of uterine proliferation ($P < 0.01$). Estradiol was significantly decreased at 1 and 2 years compared to 6-month-old females, before becoming erratic at 3 and 4 years, with no changes in testosterone across any age. Fully formed primordial follicles were observed in prenatal ovaries. Aging impacted on both OvR and growing follicle numbers ($P < 0.001$ – 0.0001). After the age of 3 years, the follicle decline rate increased more than 5-fold.

LARGE SCALE DATA: N/A.

LIMITATIONS, REASONS FOR CAUTION: This is a descriptive study in a novel research rodent whereby reagents validated for use in the spiny mouse were limited.

WIDER IMPLICATIONS OF THE FINDINGS: The gradual, rather than sudden, menopausal transition suggests that the spiny mouse is a more appropriate perimenopausal model than the current rodent models in which to examine the neuroendocrine pathways that encompass all hormonal interactions in the hypothalamic–pituitary–gonadal axis. The logistic, ethical and economic advantages of such a model may reduce our reliance on primates in menopause research and enable more thorough and invasive investigation than is possible in humans.

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Key words: menstrual cycle / rodents / menopause / ovarian reserve / uterus

Introduction

Menopause, or the menopausal transition to reproductive senescence, is the permanent arrest of reproductive capacity and simultaneous cessation of menstruation following age-related depletion of the ovarian reserve (OvR). OvR is defined as the number of quiescent primordial follicles remaining in the ovary that can be stimulated to grow and produce oocytes (Findlay et al., 2015) and provides an accurate assessment of female reproductive potential (Gnoth et al., 2018). While an age-related oocyte depletion and exponential decline in nonrenewable follicular reserve is a process in all mammalian species, humans appear to be unique in their gradual transition to reproductive senescence and prolonged sterile lifespan thereafter.

In the time leading to up to menopause (perimenopause), menstruation may become erratic. Cycles may become anovulatory with time between periods lengthening, due to considerable age-related changes in endocrinology. As follicles deplete, the ovary becomes gonadotrophin-insensitive and, in humans, estrogen levels decline significantly to almost undetectable levels (Burger, 1996). Chronological age can also be used to predict OvR and menopause. However, this is not a simple linear relationship, as the rate of follicle decline varies significantly with age and other factors and fits an exponential or biexponential regression model with follicle number plotted against age. One of the most commonly accepted models is a 'broken-stick' biexponential regression that was based on the exact follicle counts of 110 full ovaries obtained from women aged 0–51 years (Faddy and Gosden, 1996; Faddy, 2000). This model showed that in healthy women after the age of 37, follicle depletion rate increases significantly.

Over the last decades, laboratory rodents, and some other animal models, have been used extensively to aid human reproductive research (Bellino and Wise, 2003). However, no small animal model that accurately reflects OvR decline, hormonal changes, menopausal timing and onset, as well as reproductive senescence and postmenopausal longevity, has been identified for human menopausal research (Bellino and Wise, 2003; Maffucci et al., 2006).

Our recent discovery that the spiny mouse (*Acomys cahirinus*) menstruates (Bellofiore et al., 2017) naturally suggests this species may provide a more refined rodent model of menopause with better translation to the human condition. The spiny mouse, a desert-dwelling species with few captive research colonies worldwide, has been recognized in skin regenerative research (Seifert et al., 2012), *in-utero* organogenesis (Brunjes, 1990; Dickinson et al., 2005), and for its primate-like secretion of dehydroepiandrosterone (DHEA) and cortisol (Quinn et al., 2013). We have demonstrated similarities with humans in the morphological characteristics of menstrual shedding and repair in sexually mature *Acomys* females.

The aim of this study was to examine the effects of aging on menstruation and reproductive potential of female spiny mice by assessing functional and histological changes to the menstrual cycle of the aging spiny mouse, and to determine whether a perimenopause transition, analogous to that in women, occurs in this rodent. Here, we describe morphological changes during the reproductive lifespan of female spiny mice, focusing particularly on OvR. We show that the spiny mouse has a longer reproductive capacity than the laboratory mouse that is distinguished

by a gradual decline in menstrual events, follicular reserve and uterine and ovarian mass and function. These changes are associated with the decline and exhaustion of follicles in the ovary, and closely resemble the events surrounding the natural end of fertility in women.

Materials and methods

Ethics

All experimental procedures adhered with the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes. All research was approved by the Monash University/Monash Medical Centre Animal Ethics Committee (MMCA2013/11/BC, MMCB 2015/47).

Spiny mice

Virgin female spiny mice were sourced from our in-house research colony at Monash Medical Centre and maintained as previously described (Bellofiore et al., 2017). Virgin female spiny mice were assessed for physiology, stereology or endocrinology according to Table 1.

Vaginal lavage and menstrual cycle assessment

Adult female spiny mice between 3 and 6 months old have a 9-day menstrual cycle (range: 6–10 days) comprising follicular/proliferative, secretory/luteal and menstrual uterine/ovarian phases, respectively. In this study, females up to 4 years of age were subjected to daily vaginal lavage for 14 consecutive days to cover at least one full menstrual cycle. Lavage samples were prepared and stained with hematoxylin and eosin (H&E) as previously (Bellofiore et al., 2017), and cytology was assessed blindly for stage of the menstrual cycle after all samples had been obtained. Stages were designated as proliferative (rodent equivalent: proestrus and estrus), secretory (metestrus and diestrus) or menses (no rodent equivalent). Females with lavages that contained no dominant cell type, in the absence of stromal debris and/or erythrocytes were designated as 'non-cycling'. Spiny mice were weighed every 3 days to obtain an average body weight. Females were deemed to be actively cycling if they exhibited at least one ovulation (designated by a vaginal lavage containing >90% cornified epithelial cells) and one menstruation (lavage containing primarily leukocytes, erythrocytes and endometrial stromal debris).

Euthanasia

Spiny mice were euthanized by isoflurane overdose. In the physiology and stereology cohorts, this was independent of menstrual cycle stage as this could not be controlled for in females presenting as 'non-cycling'. An additional challenge in working with the spiny mouse, compared with laboratory mice, is that the menstrual cycle of cohoused spiny mice does not synchronize naturally (Bellofiore et al., 2020). Spiny mice in the endocrinology cohort were euthanized in the same manner. Since these females were not followed across 14 days for

Table 1 Cohorts and sample sizes for assessing age-related changes to the menstrual cycle in spiny mice.

Cohort	Key questions	Age	Sample size	Analyses
Physiology	When does the spiny mouse show a decline in reproductive function?	6 months	14	Vaginal cytology, uterine and ovarian histology, macroscopic data
	How does the menstrual cycle change over time?	1 year	7	
		2 years	13	
	Are there observable differences in body and organ composition in aging females?	3 years	9	
		4 years	9	
Stereology	At what age is there a rapid decline in the ovarian reserve?	PND 35	3	Quantification of ovarian follicles
	Does this correspond to physiological changes?	DOB	5	
		D35	5	
		D100	5	
		6 months	5	
		1 year	5	
		15 months	5	
		2 years	5	
		3 years	5	
		4 years	5	
Endocrinology	Is there a decline in steroid production during aging?	6 months	9	Estradiol and testosterone assays
		1 year	9	
		2 years	9	
		3 years	9	
		4 years	9	

PND, prenatal day; DOB, day of birth; D, day.

cycle monitoring, a lavage was taken at postmortem to correlate the stage of the cycle with later hormonal analysis.

Hormonal assays

Blood was collected for hormonal analysis via cardiac puncture and left to clot at room temperature for 30 min. Serum was collected following refrigerated centrifugation and stored at -80°C for later hormone assessments. Commercially available Mouse/Rat ELISA kits were used according to manufacturer's instructions to measure estradiol and testosterone (Calbiotech, CA, USA, ESI80S-100 and TEI87S-100, respectively). Three 6-month-old male samples were measured for comparison for the testosterone assay, with spike and recovery and linearity of dilution validation steps performed for spiny mouse serum (Supplementary Table S1).

Histology

Reproductive tract fixation, processing and H&E staining for morphological assessment was performed as previously described.

Immunohistochemistry

To assess changes in uterine and follicular proliferation, tissues were stained for proliferating cell nuclear antigen (PCNA, 1:750, Abcam, CA, USA, #ab152112). Validated immunohistochemistry protocols were used (Bellofiore et al., 2018) with the following of note: Citrate antigen retrieval was performed by microwaving slides in citrate buffer (pH 6.0) on medium-high for 5 min, followed by 20 min hot buffer retrieval and blocking solution (10% normal goat serum, 2% Human serum in 0.02% tris-buffered saline-Tween20) applied for 30 min at room

temperature. Antibody or rabbit IgG (Dako, CA, USA, #X093602) as a negative control was applied overnight at 4°C . Endometrial tissues were then assessed for immunopositive proliferation (% area coverage) by region-specific tracing using ImageJ software as previously described (Bellofiore et al., 2018).

Stereological quantification of follicles

Ovaries were fixed by immersion for 24 h in a standard solution of Bouin's fixative (saturated picric acid, formaldehyde and acetic acid) and then stored in 70% ethanol, which was changed every 5–7 days, until processed for cold resin glycol methacrylate (GMA) embedding (Technovit 7100, Heraeus Kulzer GmbH, Wehrheim, Germany) according to manufacturer's instructions. Embedded ovarian samples were serially sectioned at $10\mu\text{m}$ using an Leica RM 2165 microtome (Leica Microsystems Nussloch GmbH, Nussloch, Germany) and stained with H&E.

The numbers of primordial and primary follicles were quantified using a fractionator/optical dissector technique, adapted for each age cohort of the spiny mouse ovary. This method was previously optimized in a pilot study, allowing for predetermined optimal thickness and clarity of the sections, with GMA embedding minimizing tissue deformation for more accurate follicle diameter and ovarian volume estimations (Gerrits and Horobin, 1996; Dorph-Petersen et al., 2001). The systematic random sampling method (Myers et al., 2004) was also previously optimized to provide a balance between sufficient raw follicle counts and the sample analysis time .

Follicle numbers were quantified using Olympus CAST stereological software (Olympus Denmark A/S, Albertslund, Denmark) on a PC in conjunction with an Olympus BX-51 microscope fitted with $\times 40$ oil

immersion objective (numerical aperture NA = 1.0). A systematic uniform sampling procedure ('meander sampling') was used, and the total numbers of primordial and primary follicles per ovary were estimated using a mathematical model, as described by Myers et al. (2004). Secondary and antral follicle numbers were determined by exact point-counting on serial sections. Follicles were classified as primordial if the oocyte was surrounded by a single layer of squamous granulosa cells. The presence of single or multiple layers of cuboidal granulosa cells around the oocyte gave the classifications primary or secondary follicle, respectively. Antral follicles were determined by the presence of a fluid-filled cavity (Supplementary Fig. S1).

The diameter of each secondary and antral follicle was measured at $\times 10$ magnification in the section containing the oocyte nucleus. Estimation of the total ovarian volume was made using preprogrammed calculations of the section area, multiplied by the thickness of the section and the number of sections per ovary. Mean follicular diameter was measured for primordial, primary, secondary and antral follicles (every follicle measured at the cross-section containing nucleus). To do so, two perpendicular axes were drawn through the follicle, and the mean length recorded.

Follicle decline model

In humans, due to limited data on numbers of follicles and broad biological variations, decay curves are used to describe the progressive loss of OvR. We constructed and evaluated the menstruating spiny mouse follicle decay regression using a commonly accepted model, previously validated to give plausible estimations of follicle numbers in women (Faddy et al., 1992; Faddy and Gosden, 1996; Faddy, 2000; Thilagam, 2016).

The general model states that:

$$\text{Log(NGF)} = \begin{cases} \alpha_1 + \beta_1 x, & \text{for } x \leq \text{age}_c \\ \alpha_2 + \beta_2 x, & \text{for } x > \text{age}_c \end{cases}$$

where NGF = nongrowing follicles, x —the age of the subject, age_c —critical value, α —the exponential function of initial number of follicles ($\text{Log(NGF}_{\text{initial}}) = \exp(\alpha)$), β —the rate of follicle decay (log NGF versus age), estimated by regression analysis. The critical value is an exponential period of follicular decline.

Statistical analysis

The Follicle Decline Model was fitted to the data set of logarithmically transformed follicle numbers and statistically analyzed using RStudio (V.I.I.456, RStudio, Inc.). All other data were analyzed using GraphPad Prism (Ver 6.01). Normally distributed data are presented as mean $\pm$ SD, and non-normal data as median (range). Significance ($P < 0.05$) was determined using a one-way ANOVA or Kruskal–Wallis test, respectively, between all age groups. Welch's t -test was used to compare ovarian morphometry between cohorts.

Results

The menstrual cycle is prolonged with age in spiny mice

The proportion of females presenting with cytology to indicate an active menstrual cycle were: 86% (12/14) at 6 months, 71% (5/7) at 1 year, 69% (9/13) at 2 years, 56% (5/9) at 3 years and 44% (4/9) at 4 years. In 3-year (4/9) and 4-year females (5/9), postmortem revealed health issues and/or reproductive tract deformities (Supplementary Fig. S2) which were likely correlated to noncycling in these age groups. The corresponding cytology and weight data of all females in poor health were excluded from analysis as our study aimed to follow a healthy aging reproductive process.

Overall, the total length of the menstrual cycle was significantly increased in 4-year-old females (13.0 ± 1.4 days) compared to 6 months (10.3 ± 1.3 days) and 1 year (10.3 ± 1.8 days), and a trend toward an increased cycle length ($P = 0.08$) at 3 years (12.0 ± 2.9 days). This increase was attributed to an extended proliferative phase by ~ 2 –3 days in 3- and 4-year-old females compared to 1- and 2-year-old females, with a trend ($P = 0.07$) compared to 6 months (Table II).

Mean body weight of 6-month-old females was 34.5 ± 1.8 g. This significantly increased during aging in all groups compared to females 6 months of age (Fig. 1a and b, $P < 0.001$), peaking in 3-year-old females at 41.9 ± 5.5 g ($\sim 20\%$ increase in body weight). Uterine and ovarian weights, however, declined steadily from 1 year of age in all groups (Fig. 1c and d, $P < 0.01$).

The cytology of cycling 4-year-old females demonstrates some ovulatory smears within the 14-day period of assessment, designated by samples containing only cornified epithelial cells (Fig. 2a). However, noncycling females (in all age groups) demonstrated persistent vaginal cytology similar to that of a late menstrual smear, without erythrocytes, i.e. leukocytes, some cornified epithelial cells and sporadic clusters of nucleated epithelial cells. In noncycling females, vaginal cytology does not progress to the next fertile cycle and the follicular phase, denoted by dominance of smear with nucleated epithelial cells, is not initiated (Fig. 2b).

In 6-month-old females, estradiol reached peak levels in the circulation during the proliferative phase, and nadir during menses, ranging from ~ 50 to 140 pg/ml (Fig. 1e). This pattern is similar in 1-year-old females. Circulating estradiol was significantly decreased in females at aged 1 and 2 years compared to 6-month olds (ANOVA, $P = 0.019$). No differences were significant between any age groups for testosterone. Females confirmed to have reproductive tract abnormalities or signs of ill-health at postmortem were excluded from statistical analysis ($n = 3$ for 3-year and $n = 7$ for 4-year-old females). Lack of sufficient measurements for analysis from healthy animals excluded statistical comparison for steroids in the 4-year-old females; however, the complete data are depicted in Supplementary Fig. S3.

Loss of follicles and decreased organ weight

PCNA immunohistochemistry confirmed endometrial proliferation to varying degrees in all ages (Fig. 3a–e). Proliferation in the endometrium showed specific localized expression in the luminal epithelium during menstrual shedding and simultaneous repair (Fig. 3c). Changes in

Table II Menstrual cycle comparison across aging spiny mouse females.

	6 months (n = 12)	1 year (n = 5)	2 years (n = 9)	3 years (n = 5)	4 years (n = 4)	Significance
Total cycle (days)	10.3 ± 1.3 ^{a,b,e}	10.8 ± 1.3	9.6 ± 1.7 ^{c,d}	12.0 ± 2.9 ^{b,c,e}	13.0 ± 1.4 ^{a,b,d}	a, ** b, *, ** d ** e: P = 0.08
Proliferative (days) Rodent Equivalent: Proestrus, Estrus	3.3 ± 0.5 ^c	2.1 ± 0.7 ^a	2.3 ± 1.1 ^b	3.7 ± 2.2 ^c	4.9 ± 1.8 ^{a,b}	a, b ** c: P = 0.07
Secretory (days) Rodent Equivalent: Metestrus, Diestrus	3.6 ± 0.8	3.8 ± 0.9	3.3 ± 1.0	4.0 ± 1.7	3.9 ± 1.9	ns
Menstrual	3.2 ± 1.6	3.8 ± 1.3	3.7 ± 0.9	3.9 ± 0.7	4.4 ± 1.6	ns

Superscript letters denote groups which are significantly different from each other (one-way ANOVA (total cycle) and two-way ANOVA (cycle stages)). Data are mean ± SD.
*P < 0.05;
**P < 0.01.
ns, nonsignificant.

ovarian weights were mirrored in histological assessment, in which mature antral follicles and corpora lutea were easily distinguished and linked to ovarian mass at all ages, except in females aged 4 years (Fig. 3f–i). The highest incidence of PCNA-based proliferation in the endometrial stroma (median percentage of area coverage) occurred in the youngest cohort (10.91% (5.78–26.40)), with 4-year-aged females exhibiting the lowest (0.02% (0.00–0.22)). PCNA expression was reduced significantly in 1, 2 and 4 years females compared to 6-month-old females ($P < 0.01$).

Ovarian ontogeny

Histology of the spiny mouse ovary through the age groups in this study clearly reflected the main stages of human ovarian ontogeny: primordial follicle formation, prenatal establishment of the follicle reserve, follicle activation and growth, follicle depletion at menopause (Fig. 4). Fully formed primordial follicles were observed in prenatal spiny mouse ovaries 3–4 days before birth. Secondary follicles started to grow from the day of birth, with first antral follicles developing as early as 48 h after birth. During the lifespan of the female spiny mice, the pool of primordial follicles reduced, and the ovarian volume increased slowly with an increase in the number and diameter of secondary and antral follicles (Table III). The ovaries from the oldest group (4 years) were morphologically different from those of the younger animals. Aging impacted on both OvR (primordial follicle numbers) and growing (secondary and antral) follicle numbers, which declined significantly when compared to other age groups ($P < 0.001$ – 0.0001). Primordial follicle diameters were significantly smaller in the prenatal group when compared to postnatal groups ($P < 0.0001$), as well as ovarian volume when compared to all postpubertal groups except the 3-year and 4-year groups (Table III, $P < 0.0001$). There was no correlation between NGF and either the ovarian volume or the number of antral follicles in the ovary.

Most primordial follicles were located in the ovarian cortex, with a sparse distribution in the medulla and cortico-medullary border along with growing secondary and antral follicles. Of the ovaries examined in this study, the maximum number of growing secondary follicles was observed in prepubertal ovaries on Day 30 (75 ± 6.4). After puberty,

total ovarian volume increased in association with the increase observed in the number of secondary and antral follicles (Table III).

Moreover, only one ovary out of five contained an antral follicle. Abundant vascular cortical stroma showed no clear signs of cortico-medullary separation.

Follicle decline model

When the biexponential model of follicle decline was applied to the spiny mouse follicle counts, a significant inverse correlation was found between ovarian age and NGFs (Fig. 4). The critical age was determined as 1100 days, meaning that after the age of 3 years, the follicle decline rate increased more than 5-fold.

For the two distinctive time periods before and after the critical value, mathematical equations to calculate the follicle numbers at any given age were determined. According to the model, mean predicted NGF number for the spiny mouse can be calculated using following expressions:

$$\text{Log(NGF)} = \frac{8.42 - 3.049 \times 10^{-4}x, \text{ for } x \leq 1100 \text{ days}}{8.42 - 1.633 \times 10^{-3}x, \text{ for } x > 1100 \text{ days}},$$

with standard errors for intercept = 5.25×10^{-2} , for coefficient period one (≤ 1100 days) = 1.077×10^{-4} , for coefficient period two (> 1100 days) = 8.339×10^{-5} .

Discussion

We have shown for the first time, a gradual transition to reproductive senescence in the spiny mouse which is unlike that for any other rodent, but more like human menopause. Current menopause studies outside of women use either nonmenstruating rodents or nonhuman primates. Nonhuman primates such as rhesus monkeys have been used as intact, natural models for reproductive aging given their similarities in menstrual cycle physiology, but the significant cost involved and increasing ethical concerns for the use of nonhuman primates in research are limiting factors. In humans, perimenopause lasts up to 4 years but is sudden in old world monkeys (Maffucci et al., 2006). The gradual transition relative to life expectancy (up to 6 years; Edrey et al.,

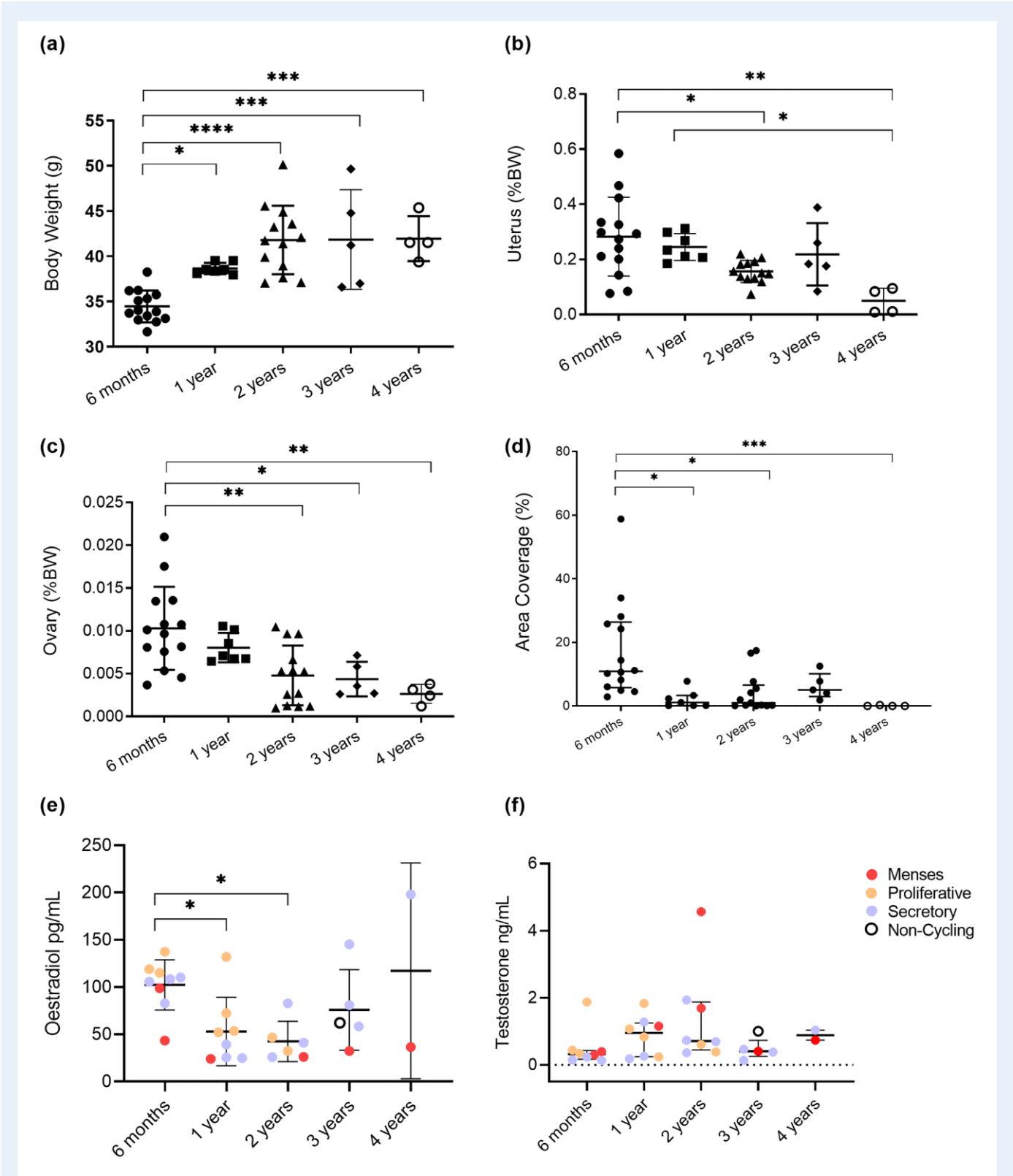


Figure 1 Morphological changes across the reproductive lifespan in female spiny mice. Changes in body (a), uterine (b), ovary (c) weight during aging in female spiny mice at 6 months to 4 years (ANOVA, %BW = percentage of body weight). (d) Changes in area coverage of positively expressed PCNA in the uterus in aging spiny mice (Kruskal–Wallis). (e–f) Changes in serum estradiol and testosterone in aging spiny mice (ANOVA). * $P < 0.05$, ** $P < 0.01$, *** $P < 0.005$ and **** $P < 0.0001$. Data are mean $\pm$ SD (a–c, e–f) and median and interquartile range (d). PCNA, proliferating cell nuclear antigen.

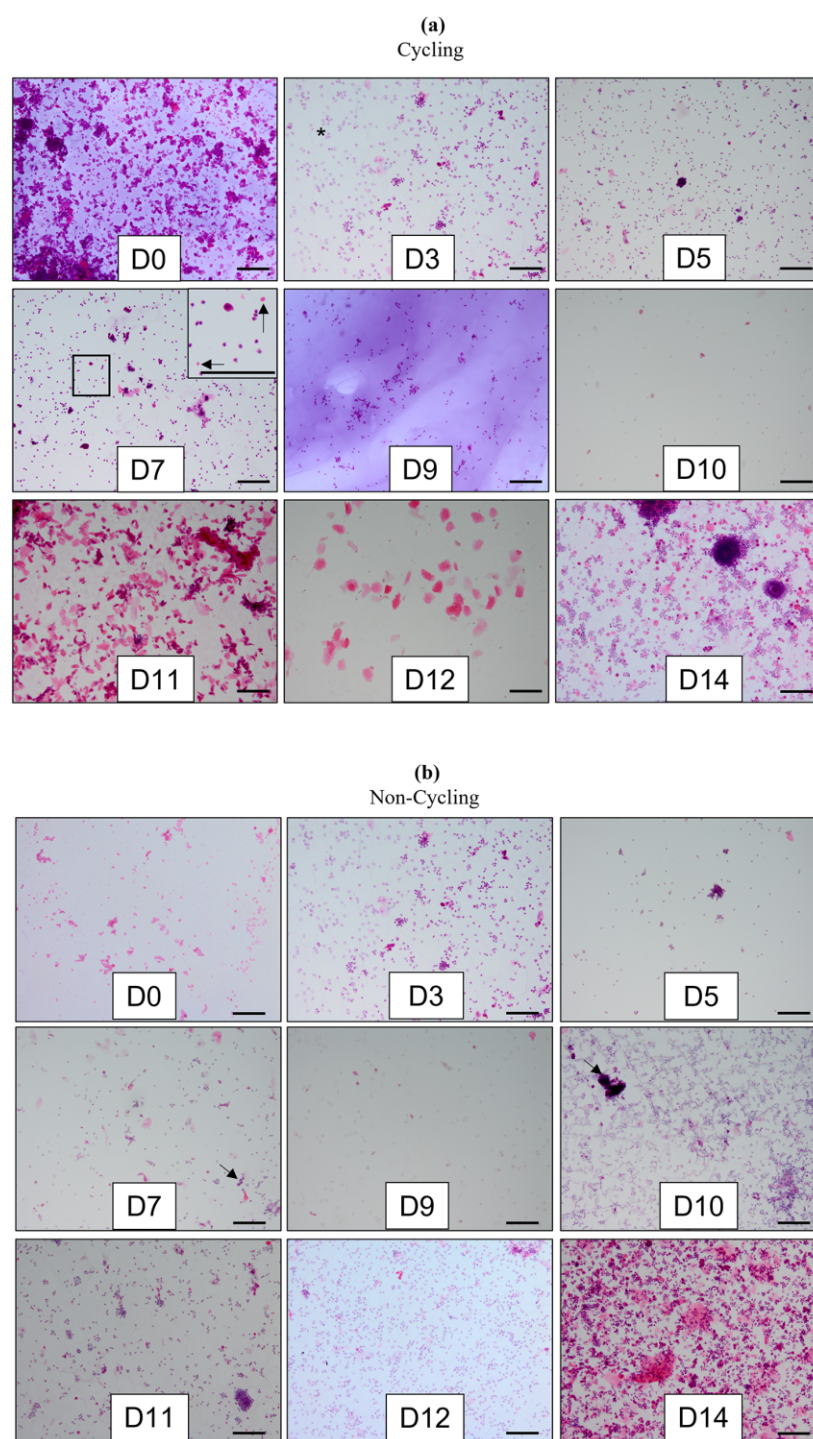


Figure 2 Cycling and noncycling vaginal cytology of spiny mice. (a) Cytology of a 4-year-old cycling female spiny mouse across 14 days. Female is in luteal phase on Day (D) 0, with late luteal phase characterized by sparse coverage of lysed, pale leukocytes (*, D3). Menstrual phase lasts from D5–D9, with extremely light bleeding indicated by low densities of red blood cells (D7, arrows). Mucus is prominent in later stages of menstrual phase (D9), leading in to early follicular phase (D10–D11). Vaginal cornification indicative of 'estrus'/ovulation is present on D12, with initial luteal phase cytology reached again on D14. Scale bars = 100 μm. (b) Cytology of a 4-year-old noncycling female spiny mouse across 14 days. Female shows cytology containing anucleated epithelial cells (D0), followed by extended periods of nonspecific cytology reminiscent of late menstrual phases, with leukocytes, some cornified and nucleated (arrows) epithelial cells (D3–D12). However, this cytology does not progress to the early follicular ('proestrus') phase and a new cycle was not initiated. D14 shows cytology containing higher proportions of cornified epithelial cells, but ovulation was not observed. Scale bars = 100 μm.

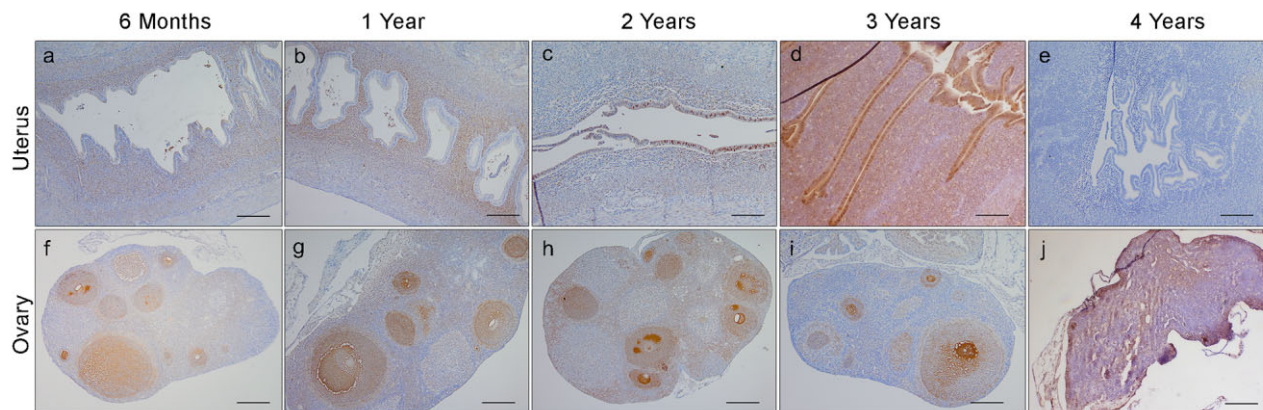


Figure 3 Morphological changes to the reproductive tract in aging female spiny mice aged 6 months to 4 years with positively stained brown cells for proliferating cell nuclear antigen (PCNA). Scale bars = 500 μ m (uterus) and 200 μ m (ovary). The endometrium has variable PCNA staining in the stromal matrix, most prominently in decidual cells and with localized staining in repairing luminal epithelium following menstruation in all groups except 4-year-old females (a–d). No staining at 4 years is evident (e). Ovaries in all age groups from 6 months to 3 years contain PCNA-positive preantral, antral and Graafian follicles, as well as corpora lutea structures (f–i). Four-year-old females show depleted reserves of follicles (j).

2012) in the spiny mouse advocates this rodent as a more affordable and relevant model.

The transition to reproductive senescence in rodents is sometimes referred to as ‘estropause’, the cessation of the estrous cycle. This differs from menopause (from Greek origins ‘menos’ meaning month, i.e. monthly period), as menopause is defined by a cessation (‘pause’) in menstruation, correlating with the reduction in follicular reserve. The standard mouse shows signs of estropause from ~9 to 15 months of age (Finch, 2014). Reproductive senescence in spiny mice is considerably longer than in this. Importantly, as in women, this transition appears gradual, rather than abrupt; approximately two-thirds of females in our captive colony retained the ability to cycle as determined by menstrual changes from vaginal lavage and the histological presence of antral follicles in females >3 years of age. Although the follicular phase showed an overall increase in length at 4 years compared to 6 months in cycling animals, we did not observe persistent vaginal cornification (PVC) in this cohort of spiny mice. Noncycling females instead demonstrate anestrus-like cytology seen in mice, with decreased cornified cells and increased leukocytic recruitment (Finch et al., 2002). Some metestrus-like samples were also observed, with higher densities of cornified epithelial cells, but these samples did not follow ovulatory phase cytology, and therefore are likely to reflect remnant estrogen production as the follicular pool is depleted, or possibly due to the increased storage of fat in older females, which is a known site of estrogen biosynthesis in postmenopausal women (Cleary and Grossmann, 2009). In addition, the degree of erythrocyte infiltration during menses in the 4-year-old cohort appeared less profound than that previously observed in young menstruating females, which may be due to declining estrogen, and an inability to support a thicker, proliferating endometrium, resulting in lighter bleeding.

Noncycling cytology is not uncommon in anxious females in our colony—those which display characteristics such as difficulty to capture and restrain, described in depth in Bellofiore et al. (2019). We have

observed this type of cytology in lavages of spiny mice subjected to high stress, such as >20 consecutive days of the handling required for sequential lavage. Indeed, stress may account for up to 25% of irregular cycling in female spiny mice under experiment (our unpublished observations). At postmortem, no reproductive tract abnormalities were found in 6-month-old controls, suggesting that stress was likely the cause of cessation of menstruation and ovulation in 2/14 females assessed. Direct transition to anestrus is less frequently observed in mice, which instead show a preference for PVC or repetitive pseudopregnancy (RPP) pathways to estropause (Finch et al., 2002; Finch, 2014). RPP is unlikely to be the cause of leukocytic dominant lavages in the spiny mouse, for several reasons. Firstly, mice and rats require pseudopregnancy induction to maintain the corpus luteum and sustain an active luteal phase to support deciduomata formation (Finn and Keen, 1963; Freeman and Neill, 1972; Terkel, 1986; Yang et al., 2009); in contrast, spiny mice exhibit an active luteal phase, functional corpus luteum and, consequently, spontaneous decidualization irrespective of pseudopregnancy. Secondly, pseudopregnant spiny mice should have functional corpora lutea in the ovaries, but these were not identified histologically. Third, the endometrium upon examination in 4-year-old females did not mirror decidualized or proliferating endometrium seen in younger spiny mice, and was negative for PCNA altogether, suggesting the equivalent of uterine atrophy seen in anestrus rodents (Finch, 2014). Finally, pseudopregnancy could not be induced in spiny mice through mechano-receptor stimulation (Bellofiore et al., 2020).

Sex steroids and their enzyme catalysts are highly conserved among mammals, though mechanisms of action and receptor ligands may differ between species (Conley and Hinshelwood, 2001; Thomas et al., 2010; Amenyogbe et al., 2020). Using a validated ELISA, a clear decline in estradiol occurred between 6 months and 2 years of age, with a hormonal milieu typical of a menstrual cycle; in which it is apparent that estradiol is elevated not only in the proliferative phase prior to ovulation but also in the secretory phase. This bimodal elevation of

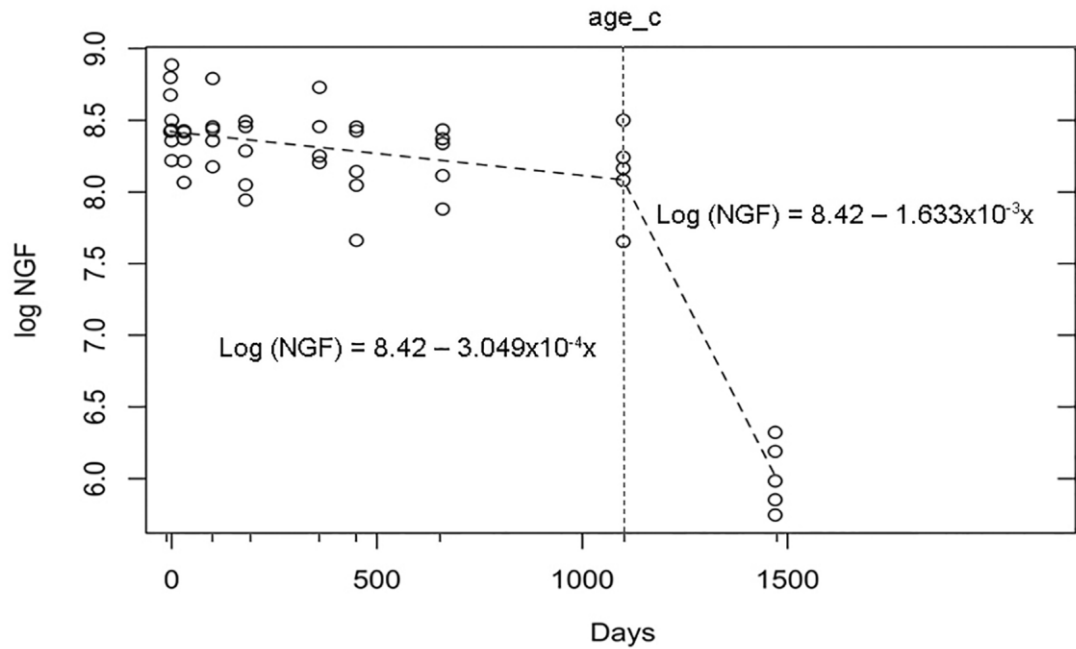


Figure 4 Biexponential model of the spiny mouse follicle decline. Nongrowing follicles (NGF) counts from prenatal Day 35 to 4-year-old mice, based on optical dissector/stereology counts from 47 ovaries, plotted against the age (days). Critical age (age_c) was determined as 1100 days, slope for period 1 (≤ 1100 days) = -3.049×10^{-4} , slope for period 2 (> 1100 days) = -1.633×10^{-3} . This model explains about 89% of variance; residual standard error: 0.2477 on 44 degrees of freedom (DF); F-statistic: 196.5 on 2 and 44 DF, P -value: $<2.2e-16$.

Table III Morphometric characteristics of the spiny mouse ovary.

	Ovary volume (mm ³)	NGF numbers	GF numbers	Primordial follicle diameter(μm)	Primary follicle diameter (μm)	Secondary follicle diameter (μm)	Antral follicle diameter (μm)
PND 35	0.086 ± 0.005	5676 ± 604	—	20.6 ± 0.2	30.1 ± 1.4	—	—
DOB	0.084 ± 0.006	4941 ± 605	21 ± 1.3	22.9 ± 0.2	32.5 ± 1.4	79.9 ± 1.5	—
D 30	1.135 ± 0.125	4058 ± 269	75 ± 6.4	23.3 ± 0.1	43.5 ± 1.3	148.4 ± 4.2	334.2 ± 8.2
D 100	1.606 ± 0.140	4738 ± 501	46 ± 15.0	22.3 ± 0.2	33.2 ± 1.7	131.3 ± 4.9	392.6 ± 14.8
6 months	2.255 ± 0.083	3899 ± 409	52 ± 12.0	24.7 ± 0.1	37.5 ± 1.2	139.5 ± 3.3	463.1 ± 20.9
1 year	1.883 ± 0.118	4493 ± 457	34 ± 2.8	24.9 ± 0.2	46.9 ± 2.5	149.1 ± 4.3	424.4 ± 33.6
15 months	2.072 ± 0.177	3588 ± 476	22 ± 4.8	23.1 ± 0.1	39.5 ± 2.0	139.0 ± 6.1	528.5 ± 27.5
2 years	1.612 ± 0.332	3816 ± 359	16 ± 3.4	24.7 ± 0.2	44.9 ± 2.6	139.7 ± 8.0	515.2 ± 40.4
3 years	1.376 ± 0.317	3513 ± 453	20 ± 3.5	23.1 ± 0.1	39.5 ± 2.0	147.8 ± 8.4	460.5 ± 42.2
4 years	0.870 ± 0.152	421 ± 45	3.6 ± 1.0	23.3 ± 0.3	31.7 ± 1.2	110.1 ± 10.0	782*

*Only one antral follicle was observed in the five ovaries examined.
Data analyzed by Welch's t-test, and results presented as mean ± SEM, n = 5 for all the samples excluding PND 35 (n = 3).
NGF, nongrowing follicle; GF, growing follicle.

estrogen is a quintessential feature of menstruating species (reviewed in Bellofiore et al. (2018)) to initiate and maintain spontaneous decidualization of the endometrium. Interestingly, despite the reduced sample size of n = 5, estrogen levels for apparently healthy 3-year-old females are similar to those of the younger controls. This is not altogether unexpected, given the wide variation in hormonal levels that is considered 'normal' in women, that female spiny mice also mimicked such natural diversity in other aspects of their menstrual physiology. Determining normal estrogen and testosterone levels in 3- to 4-year-old female spiny mice was more challenging because not every cycle

phase could be sampled due to increased tendency for other health issues. One 4-year old had a serum estradiol level of 198 pg/ml during the secretory phase with an ovary that was confirmed to have one corpus hemorrhagicum, suggesting a recent ovulation, but no other antral or preantral follicles. The elevated estradiol level may have signaled her final menstrual cycle (Supplementary Fig. S4). However, it may also be linked to a significant increase in her body mass, likely attributable to increased steroidogenesis in augmented fatty tissue in the body. Testosterone, while not significantly different between groups, showed a trend of increasing levels throughout the aging process in spiny mice (Kruskal–Wallis, $P=0.1275$). This may result from the presence of DHEA and its metabolites which are uniquely synthesized by spiny mice in the order Rodentia (Quinn et al., 2013) and cross-react in androgen assays. DHEA is an important precursor to sex steroid synthesis, so it may be that both estrogen and testosterone are using the aromatase conversion pathway to facilitate the increased levels of these steroids, as is seen in postmenopausal women (Labrie and Labrie, 2013). Further investigation here may provide a useful insight into steroid biosynthesis in the senescing spiny mouse.

While at 1–3 years of age PCNA staining is present, it is significantly reduced compared to 6-month-old females. Based on both the cytological and histological assessments, this was unexpected. Low sample size may account for this in the 1-year olds ($n=7$), as there may not have been enough females during the late proliferative phase of the menstrual cycle at the time of postmortem, or repair during the late menstrual phase may have been largely complete prior to a new fertile cycle and before sufficient ovarian estrogen from recruited follicles could induce endometrial growth.

This study has also demonstrated that the OvR declines significantly in the 4-year-aged group. Our data support the current theory that primordial follicle numbers in the ovary represent a nonrenewable pool of oocytes that is one determinant of the reproductive potential of the female during her lifespan. In women, reduction in the size of the follicle pool is linked to the perimenopause and menopause transition and negatively affected fertility (Richardson, 1993; Tal and Seifer, 2017). There is increasing debate that some mammals have capacity for postnatal germline cell renewal, having been observed in prosimian primates (Telfer, 2004). The determined capacity of these germline cells isolated from mice and human ovaries remains a source of controversy (Telfer and Anderson, 2019). While the authors cannot comment on the presence or absence of oocyte-producing germ cells in the spiny mouse, the significant decline in OvR, ovulation and menstruation observed here suggests a noncritical role in their functional menstrual cycle.

The histologically determined primordial follicle pool correlates significantly with other important OvR tests, such as anti-müllerian hormone (AMH), FSH, ultrasound and physiological age (Hansen et al., 2011). Among those, in women, AMH was proven to be an appropriate marker of reproductive aging (Van Rooij et al., 2005; Fouquet et al., 2014) with higher sensitivity than FSH for early diagnosis of premature ovarian failure (Alipour et al., 2015), but testing methods and protocols require validation for the spiny mouse, and despite several attempts, thus far we have been unsuccessful in detecting accurate AMH levels in spiny mice using commercially available enzyme-linked immunosorbent assays. Similarly, this study may have benefited from assessments of changes in gonadotrophins, especially during the menopause transition. However, as with AMH, we have previously

attempted to measure FSH and LH using multiple commercial kits (outlined in McKenna et al. (2021)) but have been unsuccessful in detecting either gonadotropin.

The cytology of only four females could be extrapolated from the original cohort of nine at 4 years, because of health issues or detection of reproductive anomalies. In our colony, tumors do not appear to discriminate against sex or age at parity; however, these tumors are generally located external to the reproductive tract, most commonly on the limbs, neck and head. The preliminary findings in this study suggest, however, that the spiny mouse may also be a candidate for studies of endometrial cancers and potentially uterine fibroids and the impact of these conditions on the menstrual cycle, as aged virgin females appear to be predisposed to these health issues. It is not known whether these issues are found in similar proportions in breeding females as this was not examined in this study. However, unlike estrous species, no female spiny mice presented with uterine pus consistent with pyometra, a significant infection of the uterus typical of older, nonspayed domesticated mammalian species. In canines, for example, persistent stimulation with estrogen at estrus and subsequent periods of progesterone dominance causes endometrial proliferation, glandular secretion and immune suppression. Repeated estrous cycles have a cumulative effect, leaving the uterus susceptible to bacterial infection (Smith, 2006). The spiny mouse does not behave as an estrous cycling species under these conditions. This potentially may be influenced by repeated endometrial immune activation immediately prior to menses, enabling cyclical immune surveillance and conditioning of the uterus.

Natural mouse and rat models also exhibit an age-related decline in reproductive function. However, although they share some features of ovarian senescence with humans, the human endocrine profiles are not mimicked. The complicating factor in these comparisons is that rats and mice have estrous cycles, as opposed to menstrual cycles as in higher-order primates. Rats show an increase in FSH prior to the onset of estropause, as in human menopause, but a moderate production of estrogen continues due to retention of a larger pool of primary follicles (Chakraborty and Gore, 2004; Maffucci et al., 2006), and in aging female mice estrogen secretion is significantly reduced but is not ablated (Van Kempen et al., 2011). For these reasons, an ovariectomized (OVX) rodent model of menopause may be preferable to study the effects of complete estrogen abolishment. However, not only is estrogen depletion sudden in rodents, rather than gradual as in humans, it is not the only hormone affected through ovariectomy. Researchers must consider the effects of altering the neuroendocrine pathways, and the interplay of other endocrines. In particular, humans have demonstrated a decline in DHEA and increase in DHEA-sulfate production from the ovaries (Davison et al., 2005; Finch, 2014). OVX rodents therefore are not authentic models for the effects of menopause as they lack ovaries and have naturally low plasma DHEA production.

In this study, 4-year-old females were the oldest animals available for the project. Initially, we conducted the study by examining most of the females from the 4-year cohort at 3.5 years and found a very gradual decline in proportion of cycling females, with over half (56%) still showing the capacity for ovulation. From this observation, we decided to allow these females to age an additional 6 months and reevaluate their reproductive activity. The increased incidence of health issues and the cost of prolonged housing/husbandry ultimately restricted our ability to assess female reproductive changes beyond 4 years. In the wild, the lifespan of *Acomys* is estimated to be around 3 years, and up to

6 years in captivity (Nowak and Walker, 1999). The most recent published observation of the lifespan of the spiny mouse in our colony was made some years ago, and prior to the discovery of menstruation (Dickinson and Walker, 2007), with adults surviving up to 4 years. Based on previous findings in conjunction with our current data, the spiny mouse appears to have late onset menopause, and may spend 20% of its life postmenopausal, like chimpanzees (Bellino and Wise, 2003). However, the previous cohort of animals were maintained in slightly different conditions; diets were supplemented with fruits and vegetables high in sugar weekly (including apples and carrots), and spiny mice weighed up to 60 g at >2 years of age (Dickinson and Walker, 2007), compared to the heaviest female in the current study; 50.1 g at 2 years of age. This increased mass may contribute to obesity-associated health problems and earlier deaths. This suggests that a future comparative analysis of the process of aging and the natural life expectancy of spiny mice in our colony and other laboratory colonies globally (for further descriptions of differences in husbandry, refer to Bellofiore and Evans (2019)), maintained under these different conditions, would be a worthwhile extension of this investigation.

Our study highlights again the broad potential of the spiny mouse as a model species for research not only in reproduction and development but also into aging. It is currently the only small laboratory animal that provides a natural model for research into menopause. The gradual, rather than sudden, menopausal transition suggests that the spiny mouse is a more appropriate perimenopausal model than the current OVX rodent models in which to examine the neuroendocrine pathways that encompass all hormonal interactions in the hypothalamic–pituitary–gonadal axis. Finally, the logistic, ethical and economic advantages of such a model may reduce our reliance on old world monkeys in menopause research, stimulate the development of a new suite of commercially available reagents for the much-needed common laboratory techniques and enable more thorough and invasive investigation than is possible in humans.

Supplementary data

Supplementary data are available at *Human Reproduction* online.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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Authors' roles

N.B. designed the study, collected tissue and conducted uterine and menstrual cycle analysis, and drafted the manuscript. E.G. collected and processed ovaries, performed stereological assessment and follicle decline modeling and drafted the manuscript. P.T.-S. and B.V. contributed to concept, formation of discussion and manuscript editing.

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Conflict of interest

The authors declare no conflicts of interest.

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